

3. (a) State what is meant by alpha emission. [2]

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(b) Americium-241 has a half-life of 432 years. Determine the percentage **decrease** in the activity of the element after 30 years. [4]

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(c) A builder is asked to install a smoke detector. He researches smoke detectors and finds that the radioactive source in the detector emits alpha particles. He decides that it is too dangerous to install the detector. Discuss whether or not he has used science well in his decision making. [2]

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(d) Describe how alpha, beta and gamma radiation can be distinguished:

- by using different absorbers
- by using an electric field **or** a magnetic field.

[6 QER]

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